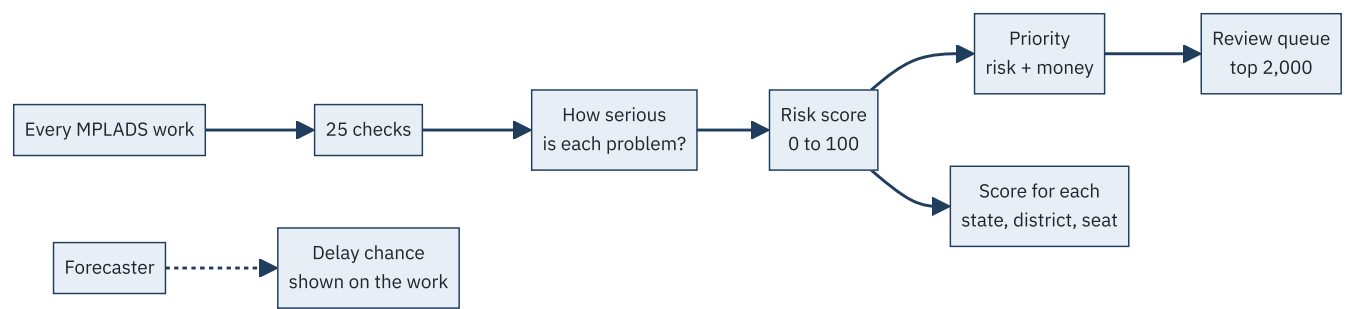


How MPLADS Review Scores Risk – in Simple Words

Aaryan Jain · 24 September 2026 · A plain-language companion to the full formulas document

The system does two separate jobs. A **rulebook** checks every work and raises a flag when something is wrong or unusual; only the rulebook can flag a work. A **forecaster** looks at a newly recommended work and estimates how likely it is to be delayed; it gives advice but never flags anything.

The big picture



Part 1: the rulebook

Step 1 – Run the checks

Every work goes through 25 checks. Each check asks one question. A few examples:

Check	The question it asks
Delay in Sanction	Did this work take much longer to be approved than similar works?
Prolonged Delay	Has it crossed a fixed time limit (for example, a year without approval)?
Excess Expenditure	Was more money paid than was sanctioned?
Unusual Cost	Does it cost far more than similar works in the same state?
Duplicate Work	Was the same work recommended twice?
Agency Concentration	Is one agency getting most of a district's money?
Completion Evidence	Was it marked complete and paid, with no proof attached?

When a check finds a problem, it creates a **finding**: a note saying what is wrong, with the evidence.

Step 2 – How serious is each problem?

Each finding gets a seriousness score from **0 to 1**, like a speedometer. Just over the line scores low; far over the line scores high.

- **0 to 0.33 = low, 0.33 to 0.67 = medium, 0.67 to 1 = high.**

- Example: the limit is 365 days. A work at 400 days is just over, so it is medium. A work at 1,100 days is three times the limit, so it is high, close to 1.

Two fairness rules adjust this:

- A problem that only "looks unusual" (a statistical check) can be at most medium on its own. It goes up a level if a *different kind* of problem is also found on the same work.
- If officers keep rejecting a type of flag, the system stops treating it as serious.

Step 3 – How much does each problem count?

The seriousness score is turned into a **strength** between 0.10 and 0.70.

$$\text{strength} = \text{weight for how serious it is} \times \text{trust in the check}$$

Trust is **1.0** for a hard rule break (for example, paid more than sanctioned) and **0.6** for "looks unusual". A plain paperwork slip counts for very little (at most 0.05).

Step 4 – The Risk score of a work (0 to 100)

Problems are grouped into kinds: time, money, rules, concentration, paperwork, data entry. Only the **strongest problem of each kind** counts, so three delay flags on one stuck work don't pile up; a delay plus a money problem do.

$$\text{Risk} = 100 \times (1 - \text{chance that none of the problems is real})$$

Example. A work has a serious delay (strength 0.60) and an unusual cost (strength 0.21). The chance neither is real is $(1 - 0.60) \times (1 - 0.21) = 0.40 \times 0.79 = 0.316$. So **Risk = $100 \times (1 - 0.316) = 68.4$** .

Step 5 – Priority: add the money

A risky ₹70 lakh work matters more than an equally risky ₹2 lakh one. Priority scales Risk by the money involved: a ₹1 lakh work counts at half its Risk, a ₹1 crore work (or more) at its full Risk, and amounts in between fall in between.

$$\text{Priority} = \text{Risk} \times \text{money factor} \quad (\text{money factor from 0.5 at ₹1 lakh to 1.0 at ₹1 crore})$$

Example. The ₹70 lakh work above has a money factor of 0.96, so **Priority = $68.4 \times 0.96 = 65.8$** . The review queue shows the 2,000 highest-priority works of ₹5 lakh or more in the current Lok Sabha.

Step 6 – Scores for states, districts, constituencies and agencies

A place's score combines **how often** its works are flagged and **how serious** those flags are:

$$\text{Place score} = \text{fair flagged rate} \times \text{average Risk, weighted by money}$$

- **Fair flagged rate.** A district with 3 works, 2 of them flagged, shouldn't look worse than a district with 3,000 works and 900 flagged. Small places are gently pulled toward the national average, like judging a new batter on more than one innings. How strong that pull is gets worked out from the data itself.
- **Only real problems count.** A work whose only issue is missing paperwork is not counted as flagged in these scores.

Part 2: the forecaster (delay prediction)

On the day an MP recommends a work, the forecaster estimates the **chance it will be seriously delayed**.

What it learned from. 90,838 past works from the 17th Lok Sabha whose outcome is known. A work counts as "delayed" if approval took more than 382 days or finishing took more than 702 days (the slowest 10%), or if it has already waited longer than that.

What it looks at, all known on the day of recommendation:

- where the work is (state, district) and what kind of work it is
- how much it costs, and how that compares with similar works
- how many works came in the same recommendation letter
- how busy the MP and the district office were in the weeks before

What it gives. A chance from 0% to 100%, and a tier:

Tier	Who gets it	How often they were really delayed
High	the riskiest 10% of works	about 7 in 10
Medium	the next 30%	about 5 in 10
Low	the other 60%	about 2 in 10

On average about 36% of works are delayed, so a High tier is roughly twice as reliable a warning as a guess. Each prediction also lists up to three reasons, for example "Recommended amount is in the top 10% of all works".

How it was tested. It was trained on older works and tested on newer ones it had never seen, which is how it will be used. It scored 0.73 on a scale where 0.5 is a coin toss and 1.0 is perfect.

Part 3: the "odd one out" score

The system also measures how unusual each work looks compared with works at the same stage (just recommended, in progress, or finished) and of similar cost. A score of 0.95 means it looks more unusual than 95% of its peers. This score only reorders works that already have the same priority in the queue; it never adds or removes a flag.

In one line

Checks find problems → each problem gets a seriousness score → problems combine into a 0–100 Risk → money turns Risk into Priority → the highest-priority works go to officers first. Separately, the forecaster warns which new works are likely to run late.